

1. DATA

Real-IAD

30 products, C1 camera
~150K images + masks

Gemini-2.5-Flash

trace generation
(img + red-mask + ref.)

AnomalyThink-6K

stratified C1 SFT split
(no second-model filter)

2. SUPERVISED FINE-TUNING

SFT four-factor grid

3B/7B · frozen/unfrozen · 6K/15K
best: 7B-frozen-6K = **80.16%**

Teacher ablation → Arm C

self-distill + STaR + Saunders
82.80% / 72.07% (SFT only)
= *best model in thesis*

3. GRPO (and its limit)

GRPO

4-component reward
(fmt + acc + type + loc), $G=4$

on *weak* SFT-Iter1 init:
→ 82.73 / 70.39 (lift)

on *strong* Arm-C init: **no gain**, ~2 pp
below init while reward rises —
holds for all 3 advantage estimators

4. EVALUATION

Evaluation (MMAD)

DS-MVtec · VisA
+ held-out Real-IAD

Best: Arm-C SFT

82.80% / 72.07%
vs. IAD-R1 (our harness)
81.92% / 71.34%